

Valve Structures and Control Valves

Learning material for Field Devices and Instrumentation

JAMK University of Applied Sciences



Learning objectives

After studying this material, you can

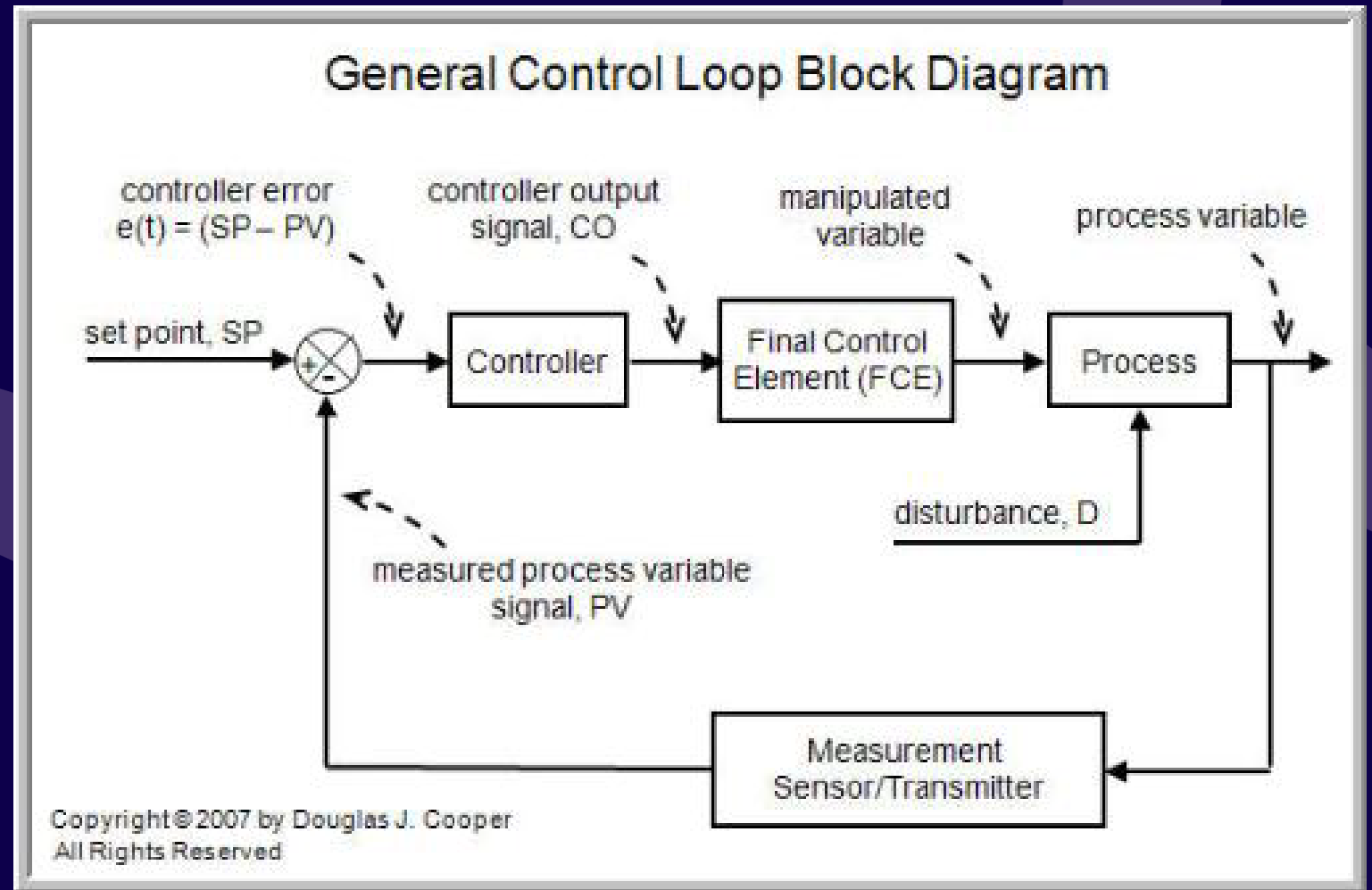
- Explain control valves in loops, including gain, sizing and capacity
- Classify valves by purpose, motion and structure
- Identify the parts of a valve, actuator and complete assembly
- Compare common rotary, linear and special-purpose valve types
- Compare actuator types, mechanisms, acting modes and fail-safe behavior
- Select positioners and accessories for control, monitoring, safety and diagnostics

Control valves in control loops

The valve is the final control element

Core idea

- Measurement is compared with the setpoint
- Controller calculates the correction
- Actuator (Control Valve) changes the process



Why valve performance matters

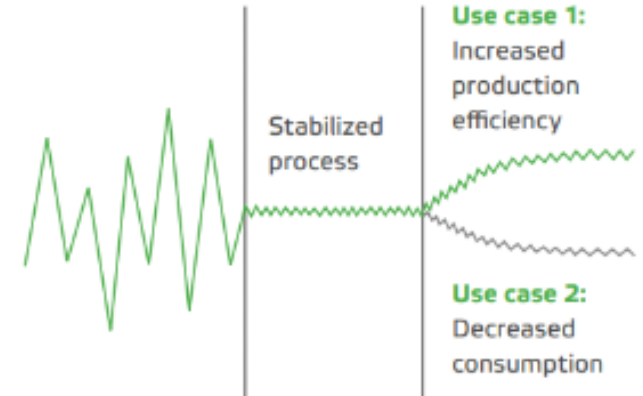
Poor control can reduce capacity and increase consumption

Control loop impact

- [Poorly functioning loops can prevent full capacity](#)
- Good performance supports operation closer to optimal setpoints
- Reliable field equipment lowers interruption risk

Improving performance based on real-time data

Over a mill's life cycle, optimizing process performance based on real-time data can yield tangible results in terms of profitability. More stable flow characteristics ensure process efficiency and the even and predictable quality of the output. Accurate process control data also enables the effective planning of predictive maintenance operations, reducing the risks of valve failure and expensive unplanned shutdowns.



Reflection question: Why does the valve assembly have such a large effect on the accuracy of the whole control loop?

Valve classification

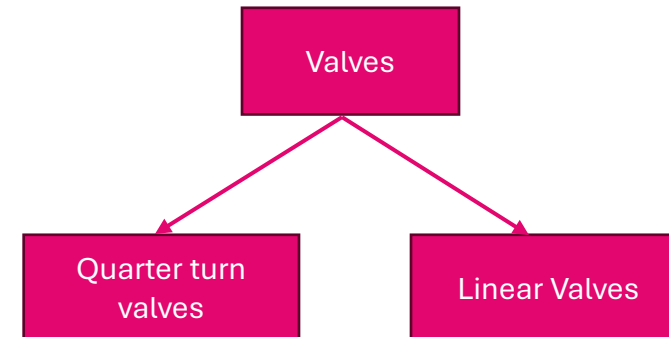
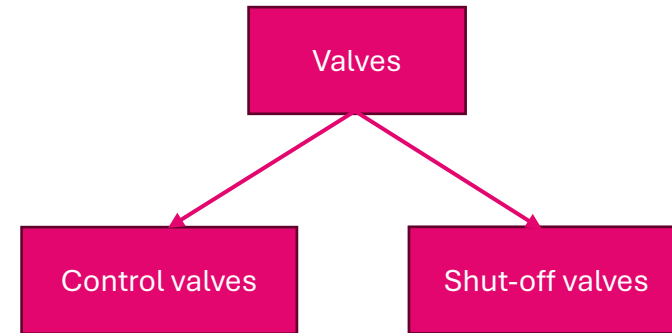
- Purpose and movement type

By purpose

- Shut-off valves: fully open or closed
- Control valves: continuously regulate flow

By structure

- Linear motion valves
- Quarter-turn valves



Control valve assembly

A control valve often means the complete assembly

Typical parts

- 1 Valve body and closure element
- 2 Actuator
- 3 Positioner
- Accessories when needed



Characteristics of a good control valve

Selection is a trade-off between process needs and constraints

- A good control valve should provide

- Good controllability
- Low noise and cavitation risk
- Resistance to mechanical and chemical stresses
- Sufficient capacity across running conditions
- Tightness
- Installability and maintainability

- **Important idea**

- The “best” valve is application-specific: performance depends on process conditions and constraints.



When is good shut-off more important than good control performance?

Valve gain

Valve position changes should cause predictable process changes

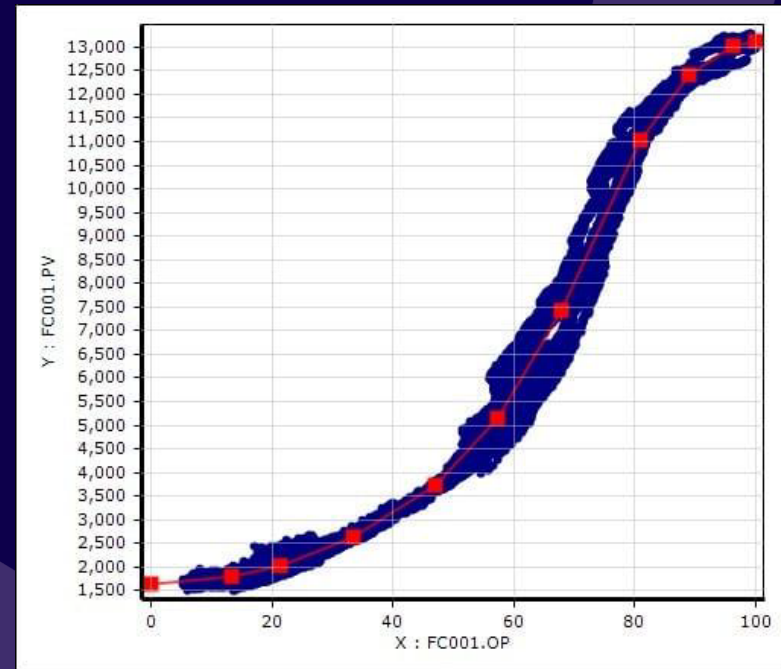
Valve gain

- Measured-variable change $\div$ valve-position change
- Example: 1% opening $\rightarrow$ 2% flow; gain = 2
- Good control needs reasonable gain across the operating range
- Valve type and design determine the characteristic—and the gain

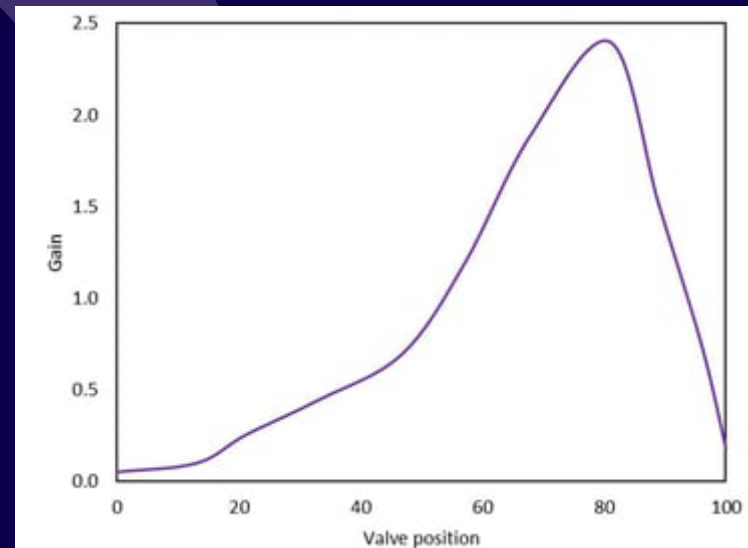


Key risk: High gain can lead to hunting: the valve and control loop oscillate.

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Valve opening (controller output) against flow (PV) to reveal the flow characteristic



Valve position against curve slope to see how gain changes with opening

Valve types

Grouped by operating motion and service

Quarter-turn valves

- Butterfly • Ball • Segment • Plug

Linear-motion valves

- Globe • Gate

Special-purpose valves

- Diaphragm • Pinch • Check
- Knife gate • Rotary plug • Safety / relief

Main components of valve

Body

- Pressure-containing structure

Closure element

- Controls flow

Seat

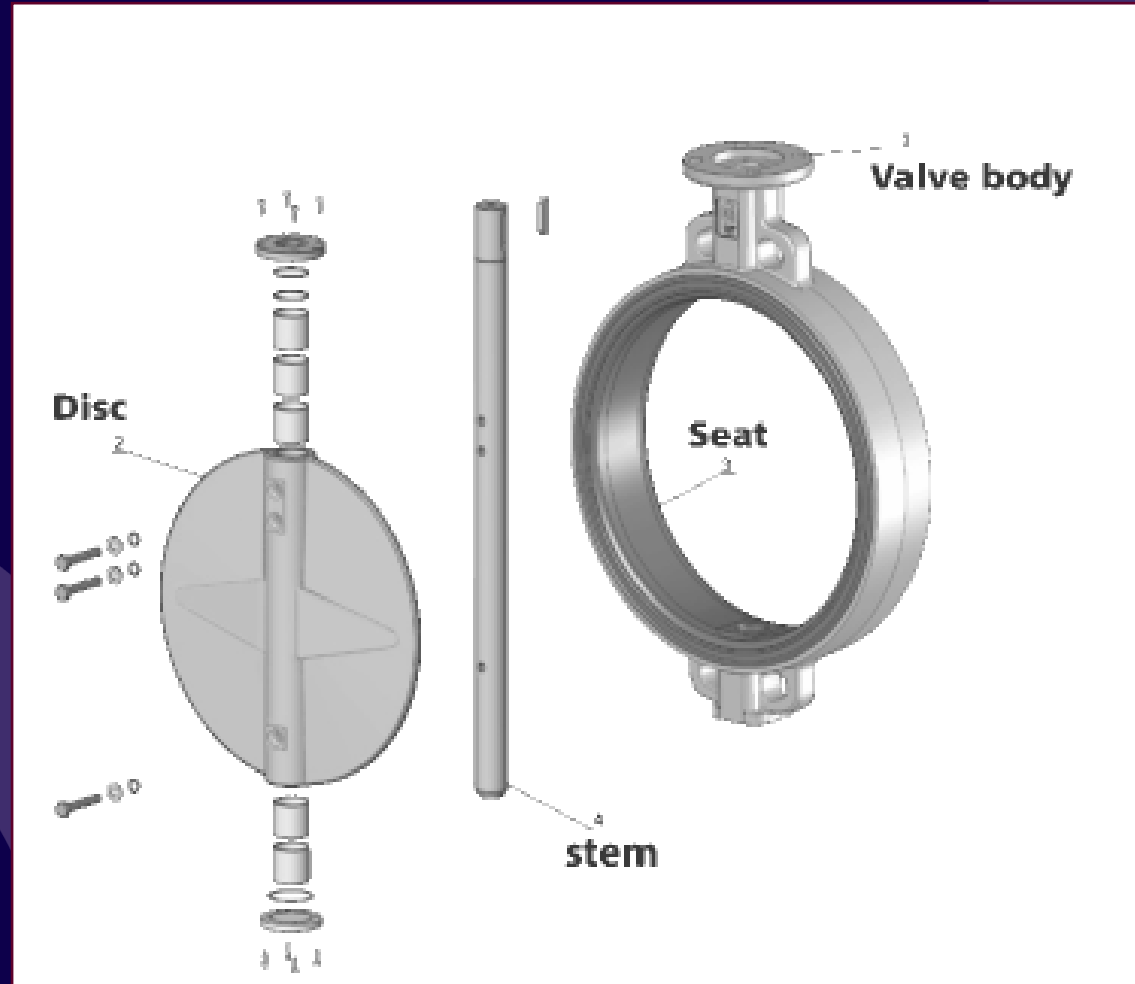
- Provides shut-off surface

Shaft / Stem

- Transfers actuator torque

Seals

- Prevent leakage



Learning activity: Use the exploded valve figures on the next slides to locate these components.

Butterfly valves

Lightweight structure, but limited control range

Advantages

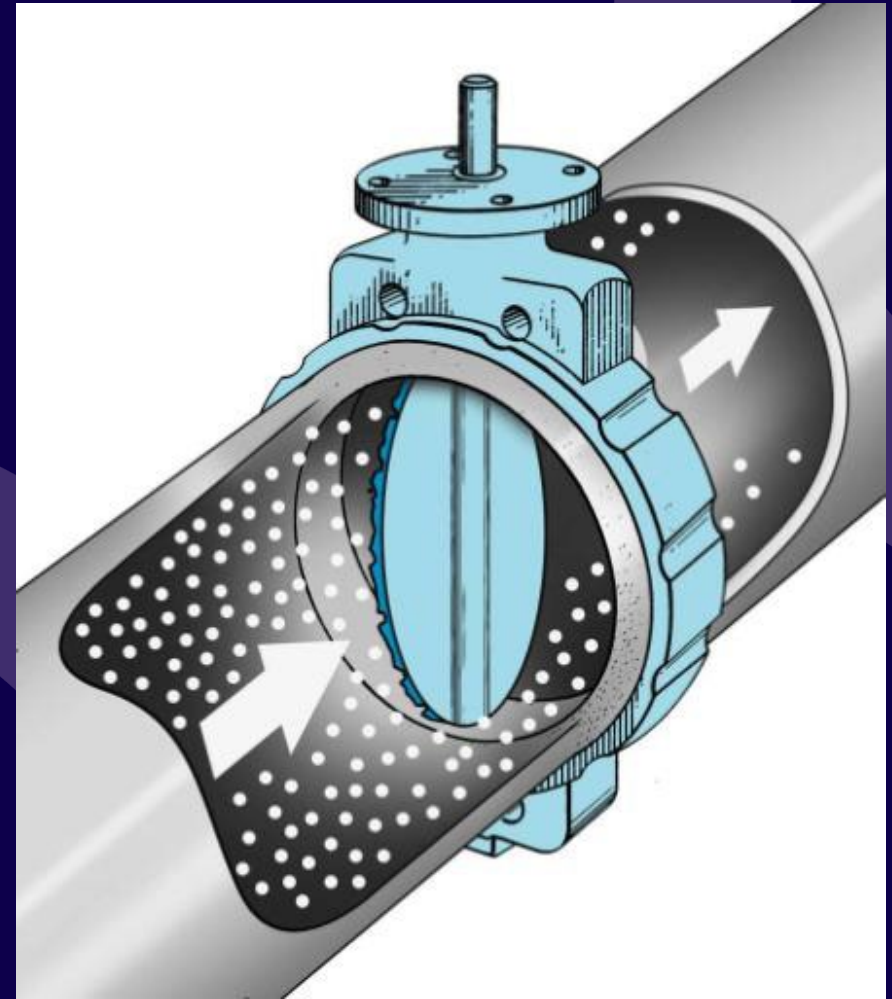
- Simple design
- Lightweight for nominal size
- Cost-effective compared with other valve types

Control limitation

- Disc remains in the flow path
- [Flow characteristics are not linear](#)

Structures

- [Triple Offset Butterfly Valve: Tri Lok | Bray](#)
- [Eccentric butterfly valve](#)



Rule of thumb: Butterfly valve useful control range is often about 25–50% valve opening.

Ball valves

Good flow characteristics and tight shut-off

Advantages

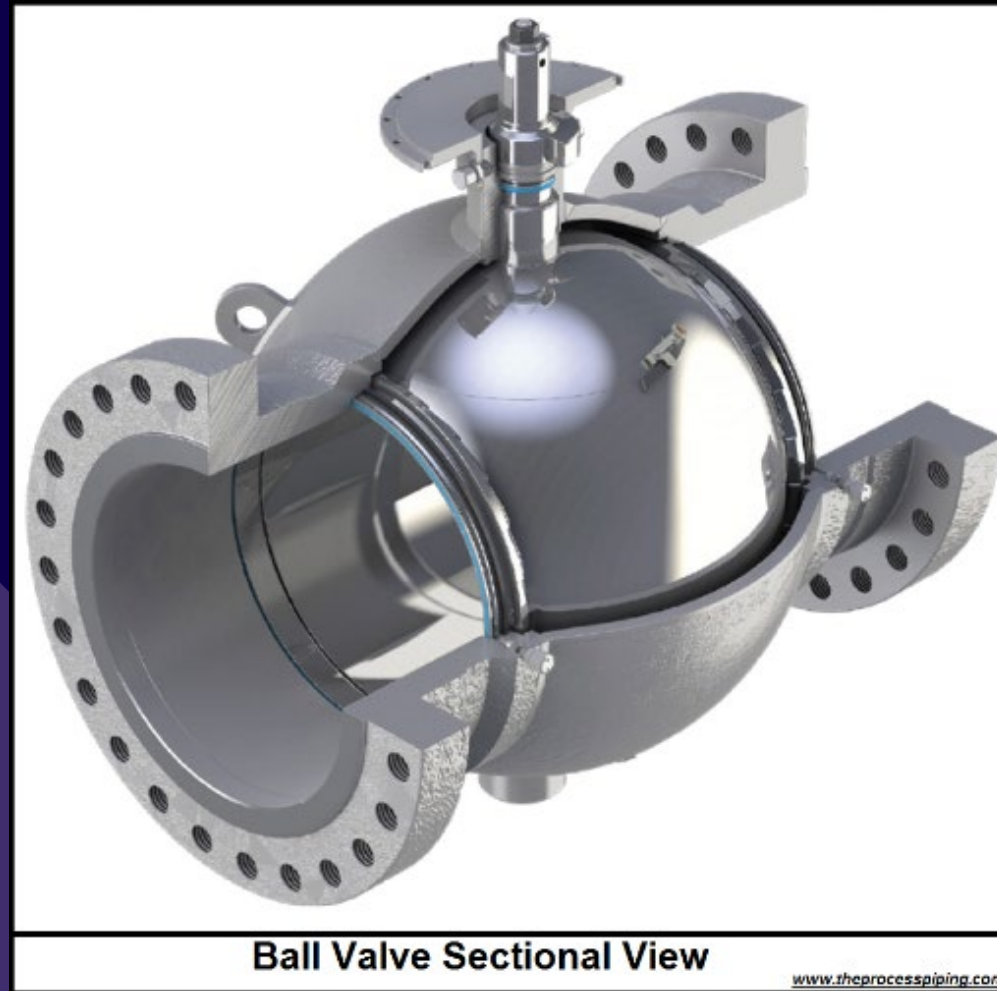
- Excellent flow capacity
- Low pressure loss when fully open
- Good shut-off capability
- Reliable general-purpose option

Possible limitations

- Massive construction for nominal size
- More structural material than some alternatives

Different structures

- [Floating ball](#)
- [Trunnion Mounted](#)



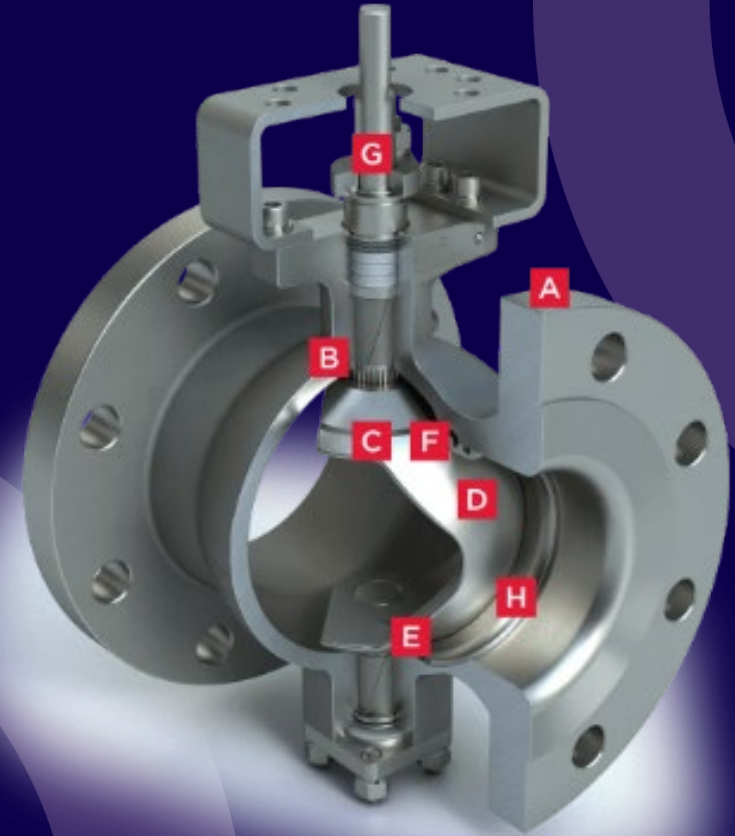
Identification task: Find the closure element, seat, shaft and seals from the exploded view.

Segment valves

Wide control range and good linearity

Why suitable for control

- Wide control range
- Linear characteristic curve
- Self-cleaning contact during rotation
- Lower friction than a ball valve
- More info: [Segmented Ball Valve Basics: Key Features and Operational Mechanisms](#)
- [Segmented Control Ball Valve Closer Look | Bray](#)



Why can a segment valve use a smaller actuator than a ball valve in some applications?

Plug valves

Quarter-turn isolation with a compact flow path

Strengths

- Fast 90° operation
- Simple, compact construction
- Tight shut-off and multi-port routing
- Well suited to frequent cycling
- Easy exchangeability of control plugs

Trade-offs

- Friction raises torque in larger sizes
- Lubricated designs may not suit clean service

Common configurations

- [Lubricated or sleeved/non-lubricated](#)
- Two-way, three-way or multi-port body

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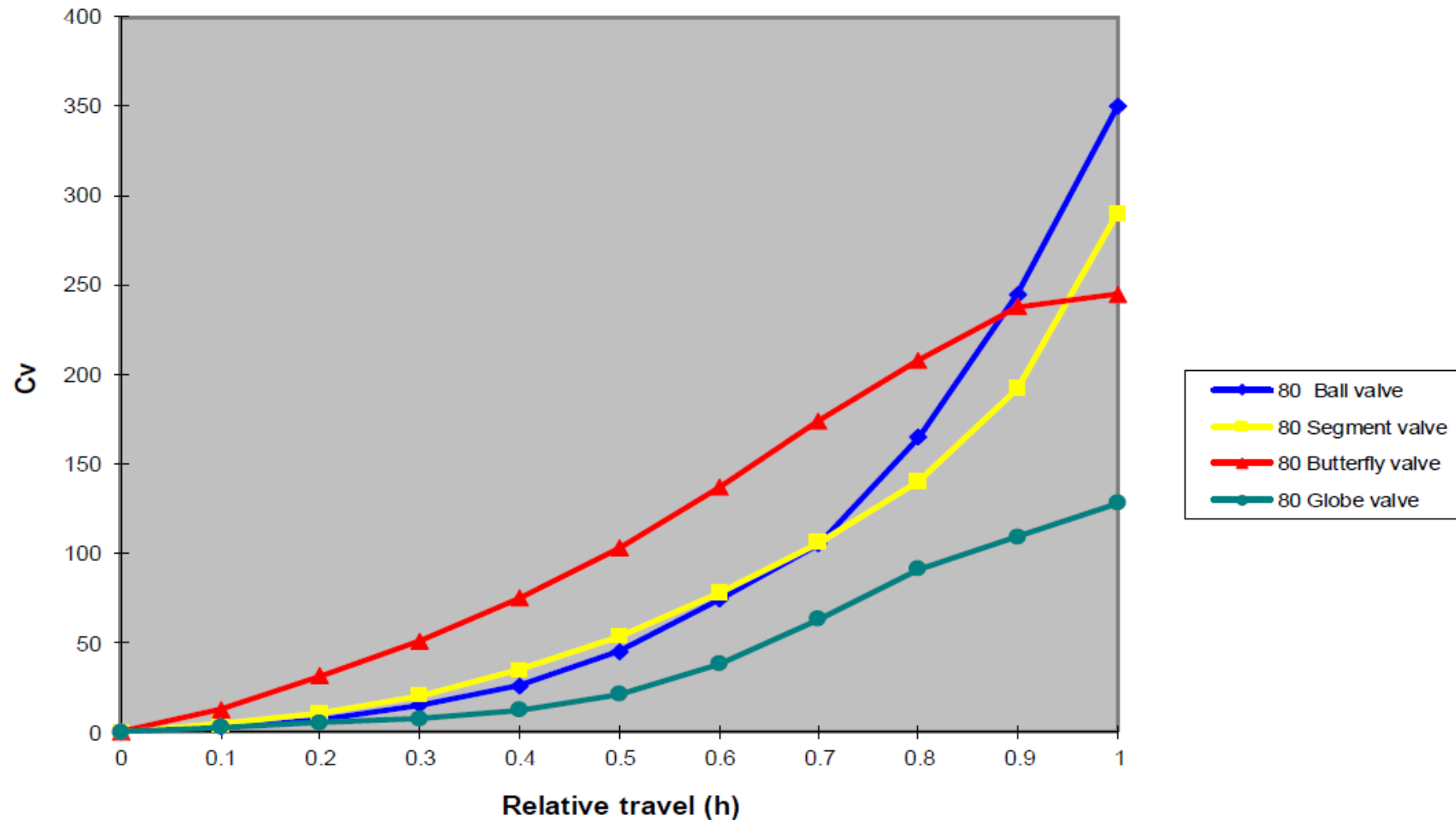
Key operating insight: Conventional plug valves isolate or divert; eccentric rotary plug variants are optimized for throttling.

Comparison of quarter-turn valve types

Use the table to support valve selection reasoning

Feature	Ball valve	Butterfly valve	Segment valve	Plug valve
Shut-off performance	Excellent	Good	Good	Excellent
Control performance	Good	Limited	Excellent	Good (eccentric)
Weight	Heavy	Light	Medium	Medium
Control range	Wide	Narrow	Very wide	Moderate
Typical use	General service	Large diameters, ON/OFF	Accurate control	Isolation / diversion

Typical capacity of DN80 control valves



Globe valves

Precise throttling through linear plug movement

Strengths

- Accurate and stable throttling
- Linear or equal-percentage trim options
- Wide rangeability for control service
- Replaceable trim supports maintenance

Trade-offs

- Higher pressure loss than rotary valves
- Larger and heavier construction
- More info: [Globe valves | Valmet](#)

Common configurations

- Straight-pattern, angle, or three-way body
- [Unbalanced](#), [balanced](#), or cage-guided trim



Key control insight: The inherent flow characteristic—and consequently valve gain—depends on the valve type and trim design.

Gate valves

Low pressure loss for isolation—not throttling

Strengths

- Straight-through, full-bore flow path
- Very low pressure loss when fully open
- Suitable for large line sizes
- Good isolation in clean environment

Trade-offs

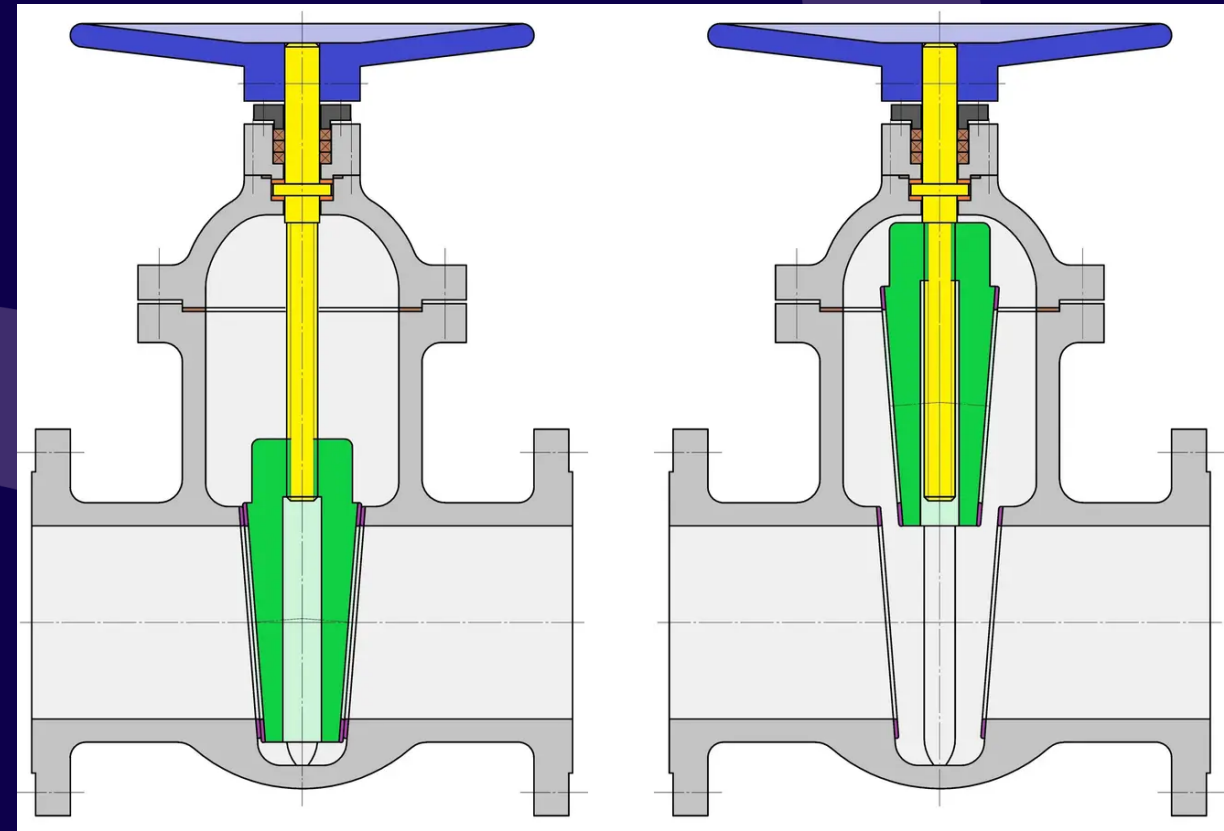
- Slow, multi-turn operation
- Partial opening can damage gate and seats
- More info: Gate valves | Spirax Sarco

Common configurations

Wedge or parallel-slide gate

[Rising-stem or non-rising-stem design](#)

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Key operating insight: Operate fully open or fully closed; throttling can cause vibration, erosion and unstable flow.

Special-purpose valves

Selection follows the medium, flow direction, isolation duty and safety function, mostly for ON/OFF applications

Diaphragm valve

Separates the actuator from the medium; useful for hygienic, corrosive or contaminated service.

Pinch valve

A flexible sleeve handles abrasive slurries, powders and solids with a full-bore path.

Check valve

Self-acting non-return protection prevents reverse flow without an external actuator.

Knife gate valve

A thin gate shears through slurry or fibrous media; primarily an isolation valve.

Rotary plug valve

A control-oriented plug variant; eccentric motion reduces seat wear and operating torque.

Safety & relief valves

Open automatically to protect a pressurized system from exceeding its safe pressure.



Linked examples: Open each valve name for a manufacturer or learning example.

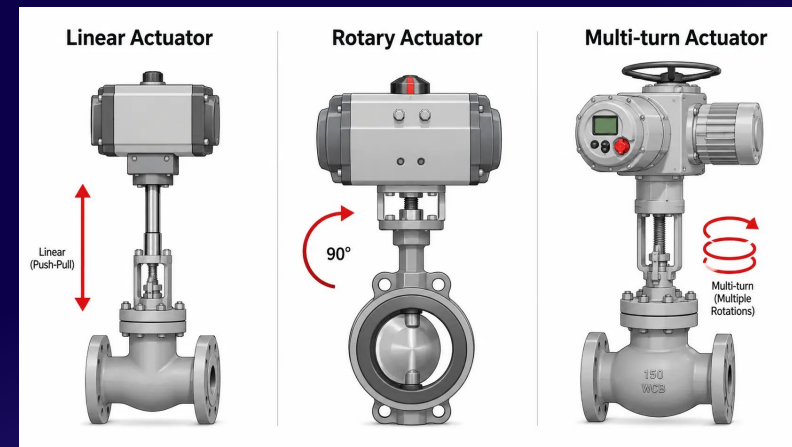
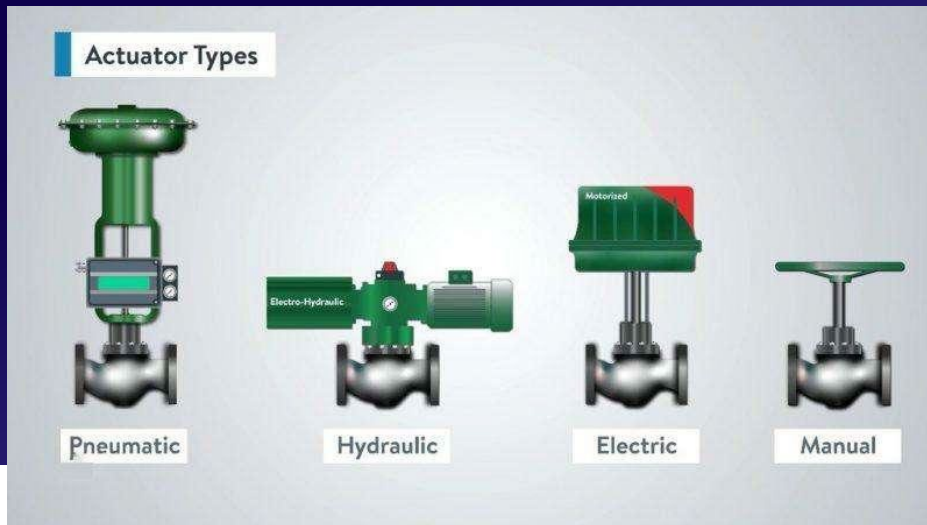
Actuators

A valve actuator converts an energy source into controlled mechanical movement

Core principle

Energy input → mechanical movement → valve position

The actuator moves the closure element and matches linear or rotary valve motion.



Movement

LINEAR — straight-line stem travel

ROTARY — shaft rotation, multiturn or 90°

After this section, you can...

Classify actuators by energy source

pneumatic,

electric hydraulic

Identify key pneumatic structures

Compare diaphragm, cylinder, rack-and-pinion and scotch-yoke designs

Explain acting mode and fail-safe behavior

Select an actuator for a typical valve duty

Classification by energy source

Energy availability, force, speed and failure behavior shape the choice

Actuator type	Energy source	Typical strengths	Typical limitations	Typical applications
Pneumatic	Compressed air	Fast, robust and easy to configure for spring fail-safe duty	Requires clean, dry air; moderate power density	Process control, on–off and ESD valves
Electric	Electrical energy	No air network; integrated control, feedback and diagnostics	Often slower; fail-safe needs a specific design or backup	Gate, globe, ball and butterfly valves
Hydraulic	Pressurized fluid	Very high force or torque; compact power density	Needs pressure source/HPU; leakage and maintenance risk	Large valves, offshore and high-pressure duty

Selection lens: Start with valve motion and required thrust/torque—then check utilities, duty cycle, environment and safe state.

Pneumatic actuators

Compressed air creates linear or quarter-turn movement

AIR PRESSURE → MOTION

Why pneumatic?

Simple, robust and easy to combine with a positioner for on-off or modulating control.

By movement

- Linear actuators
- Quarter-turn actuators

By acting mode

- Single-acting
- Double-acting

Typical pairing

- Globe → linear
- Ball/butterfly → rotary

Control options

- Modulating service (Control)
- On-off service

More info: [Neles pneumatic actuators](#)

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Pneumatic actuators are the most widely used actuators in industry.

Diaphragm actuator

Air pressure on a flexible diaphragm creates linear stem travel

AIR PRESSURE $\downarrow$ STEM TRAVEL $\downarrow$

Operating principle

- Air pressure deflects a flexible diaphragm. A spring commonly opposes the motion and returns the valve when pressure is removed.

Typical use

- Globe valves
- Linear control valves

Advantages

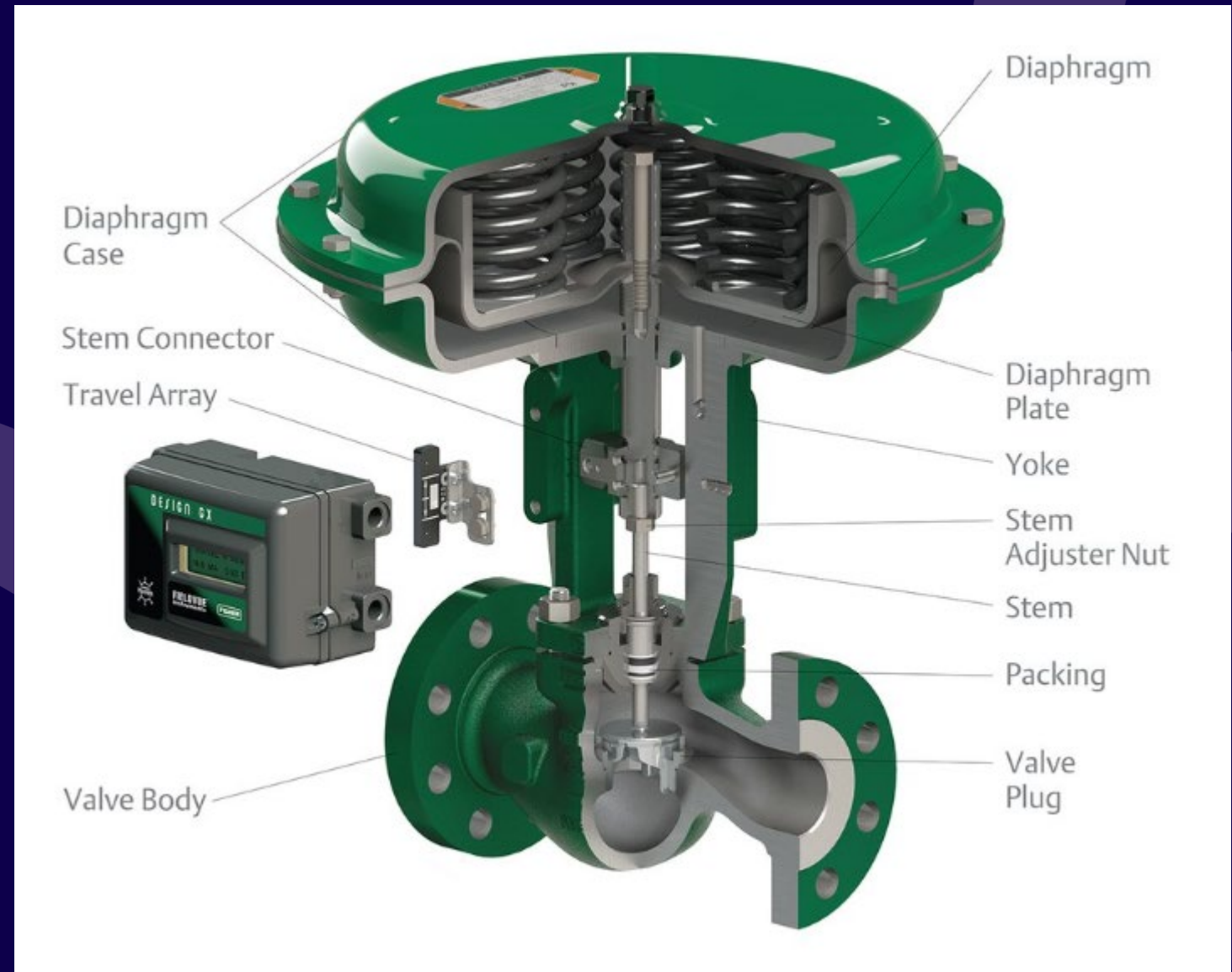
- Simple and reliable
- Smooth modulating control
- Easy spring fail-safe

Limitations & Structures

Limited force and travel

Primarily linear motion

Rotary motion



Cylinder / piston actuator

Air pressure moves a piston to produce higher thrust or torque

AIR PRESSURE → PISTON TRAVEL → ROTARY MOTION

Operating principle

Compressed air moves a piston inside a cylinder. The piston rod drives a valve directly or through a rotary conversion mechanism.

Typical use

- Ball, butterfly and segment valves
- Large linear valves

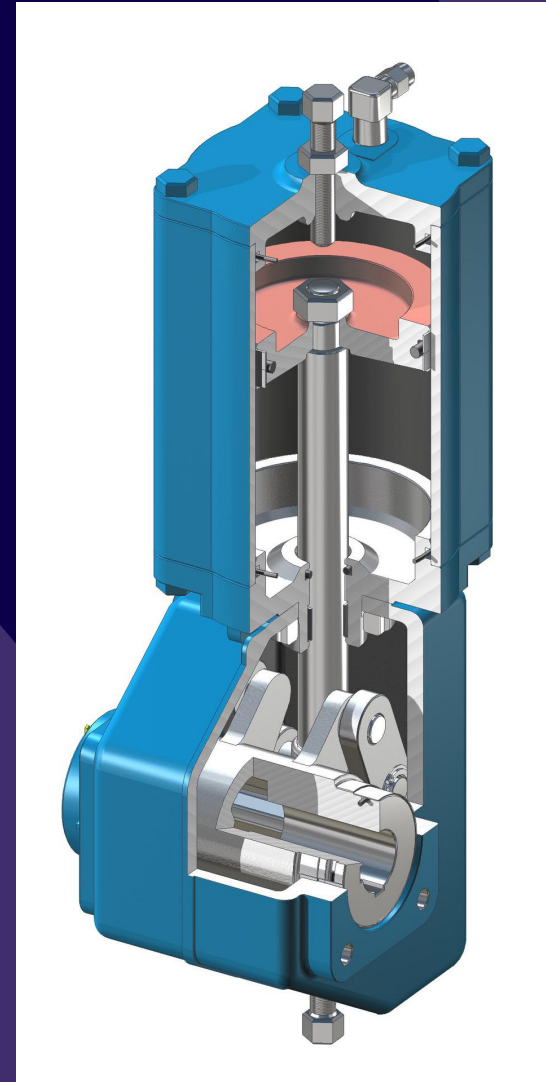
Advantages

- [Higher force than diaphragm designs](#)
- Single- or double-acting

Limitations

- More components and seals
- Quarter-turn duty needs a mechanism

More info: [Fisher 685 piston actuators](#)



Learning activity: Compare torques and torque curves to Diaphragm actuators

Rack and pinion actuator

Opposed racks convert piston travel into a compact 90° rotation

PISTONS ↔ RACKS ↔ PINION ↻

Operating principle

Pistons move toothed racks. The racks rotate a central pinion connected to the valve shaft—typically through 90 degrees.

Typical use

- Ball and butterfly valves
- Small/medium quarter-turn valves

Advantages

- Compact and simple
- Common for on-off duty

Limitations

- Torque curve may not match large valves
- Lower end-position torque than many other designs



Scotch yoke actuator

A slotted yoke converts piston travel into high-torque quarter-turn motion

PISTON TRAVEL → YOKE ROTATION

Operating principle

A piston rod drives a slotted yoke connected to the output shaft. Torque varies across the stroke and can be high near the end positions.

Typical use

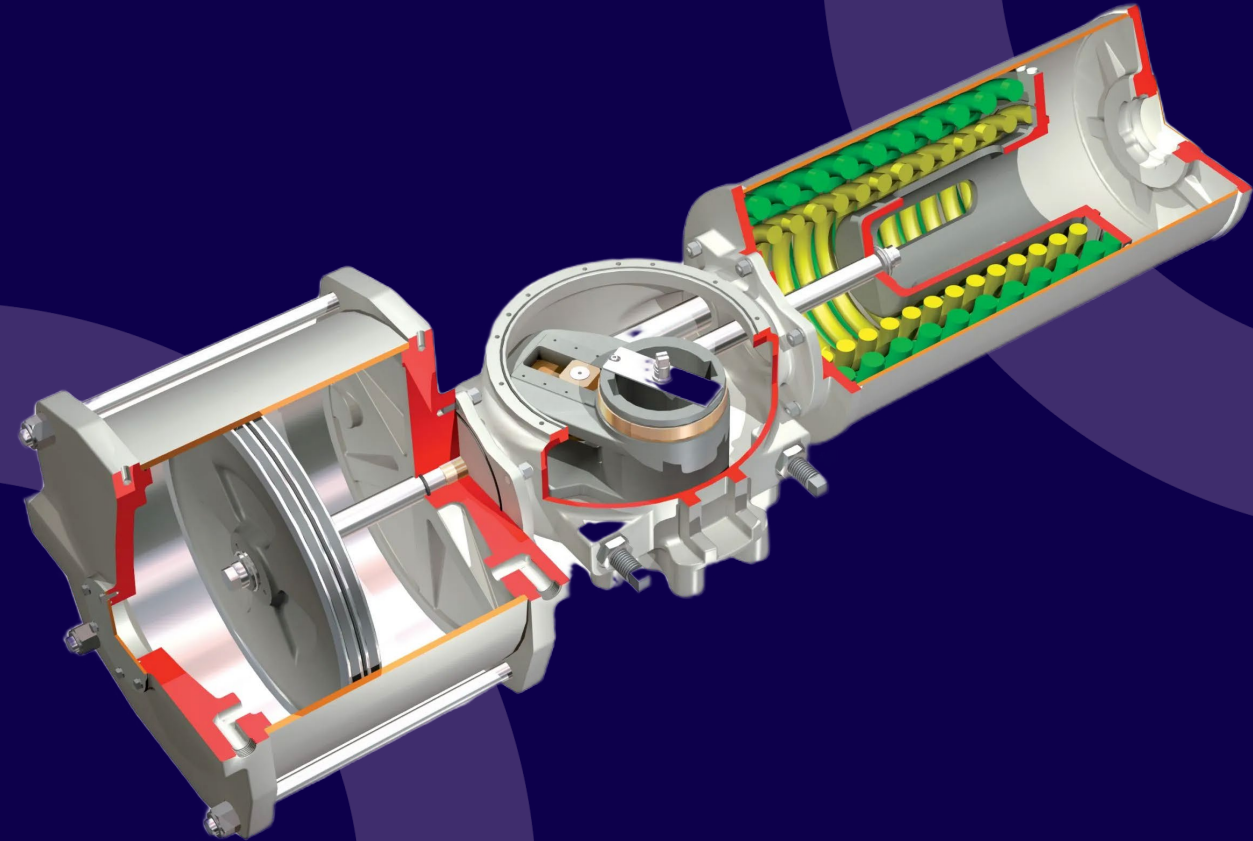
- Large ball and segment valves
- Emergency shutdown valves

Advantages

- High torque capability
- Favorable end-position torque

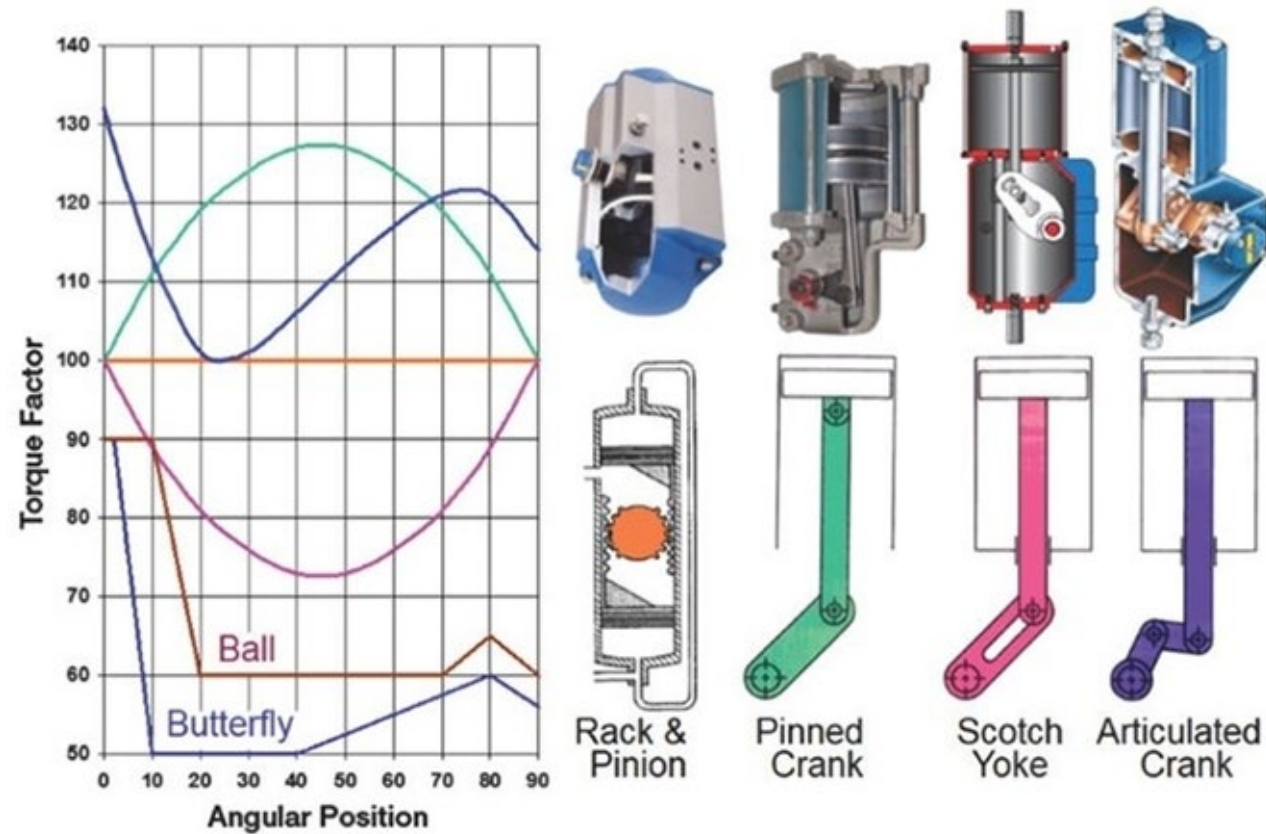
Limitations

- Larger and more complex
- Costly for small valves



Note: Useful when the valve requires high torque near fully closed or fully open.

Compare actuator torque curves



Fail-safe operation

The safe valve position is a process decision—not a universal default

**POWERED STATE → LOSS OF AIR/POWER →
Failure response**

FAIL CLOSE

Valve closes on loss of energy. Example: fuel isolation.

FAIL OPEN

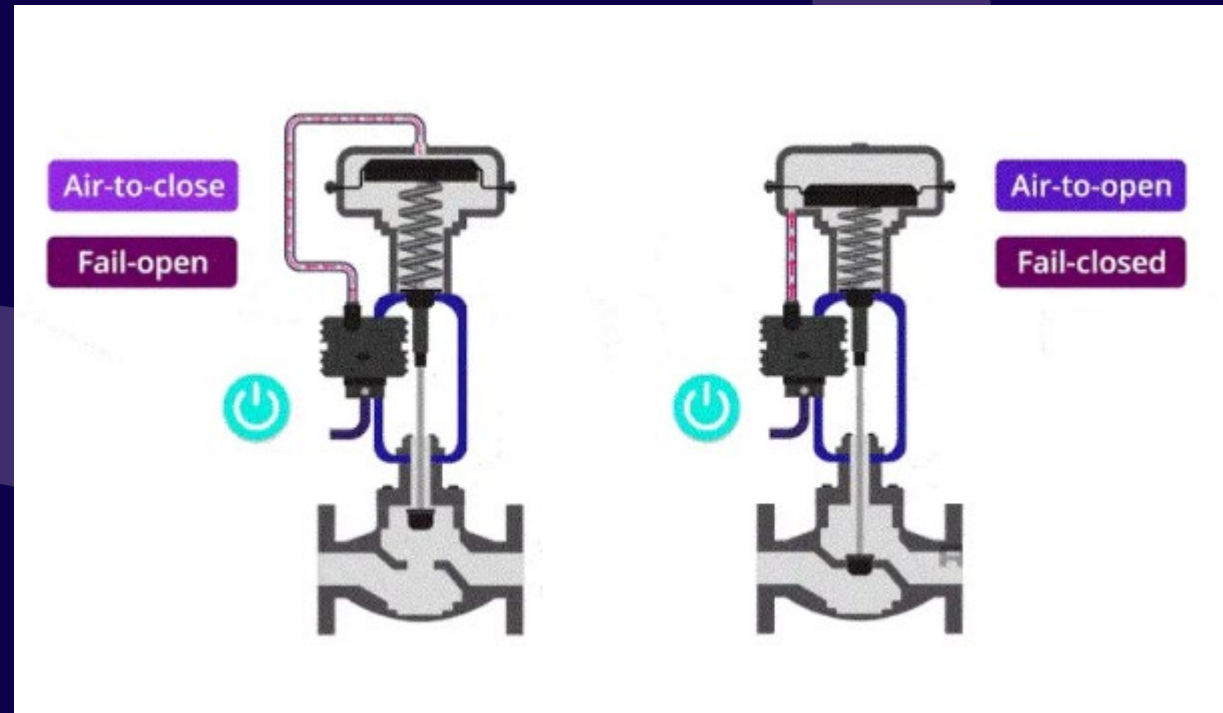
Valve opens on loss of energy. Example: essential cooling flow.

FAIL LAST

Valve is held near its last position.

Choose from the hazard analysis

- Loss of air, power or control signal may produce different outcomes
- Verify the complete valve–actuator–positioner–accessory assembly



Student question: What should the valve do if the actuator loses its energy supply—and why?

Single-acting and double-acting actuators

Acting mode determines how motion is produced—and what happens when air is lost

Feature	Single-acting	Double-acting
Air connection	Air powers one direction	Air powers both directions
Return movement	Spring or process force returns the actuator	Air pressure drives the return stroke
Fail-safe behavior	Can provide a defined spring-return safe state	Not inherent; needs added spring, accumulator or lock-up system
Typical use	Emergency shutdown and safety-related duty	Frequent cycling and modulating duty in both directions
Complexity	Spring sizing and compressed spring energy	Two air paths; no return spring in the basic design

Check your understanding: Which design provides a mechanical return movement when the air supply is removed?

Electric actuators

An electric motor and gearbox provide multi-turn or quarter-turn motion

MOTOR → GEARBOX → OUTPUT

Main types

- Multi-turn
- [Quarter-turn](#)
- Smart / networked

Advantages

- No compressed-air supply required
- Precise positioning and easy automation integration

Limitations

- Typically slower than pneumatic actuation
- Fail-safe action requires additional design or backup power

Typical valves

Gate, globe, ball, butterfly etc

More info: [AUMA multi-turn actuators](#)

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Selection note: Useful where compressed air is unavailable.

Hydraulic actuators

Pressurized fluid delivers very high force or torque

HYDRAULIC PRESSURE → HIGH OUTPUT

Operating principle

Pressurized hydraulic fluid acts on a piston. The resulting linear force can drive a valve directly or through a rotary mechanism.

Typical use

- Large valves
- High-pressure systems
- Offshore / power / ESD

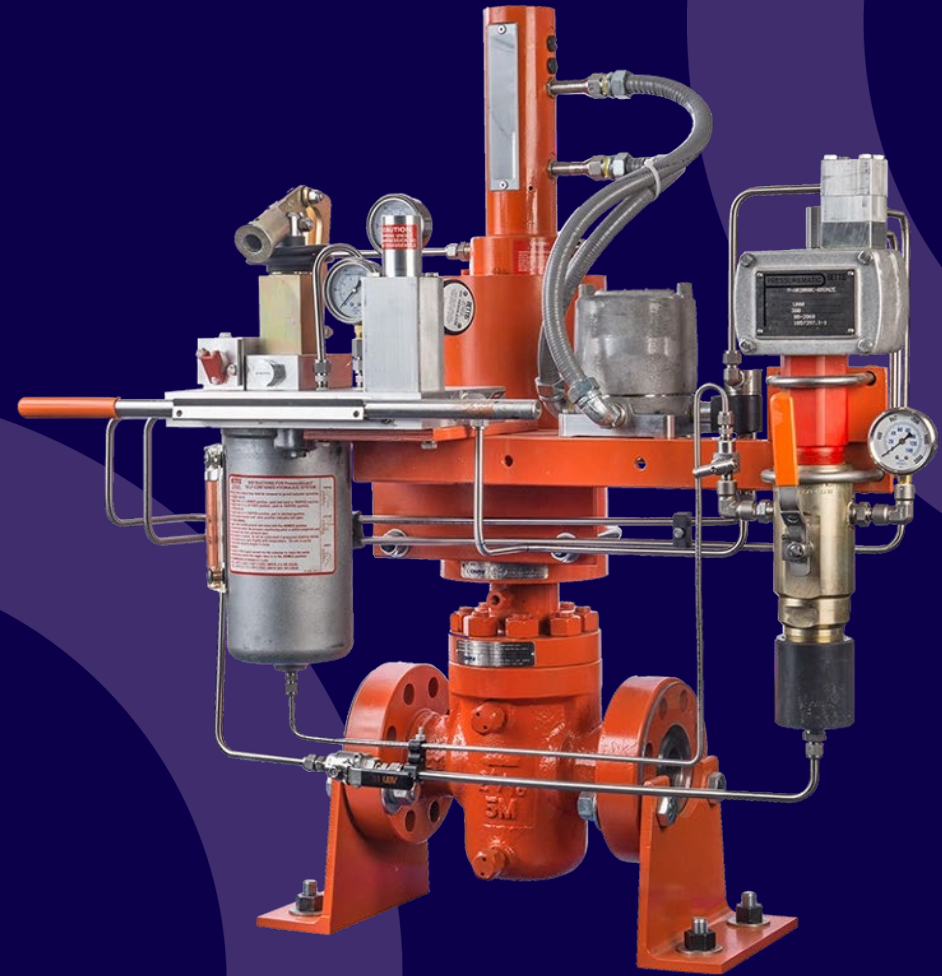
Advantages

- Very high force
- Very high torque
- Demanding environments

Limitations

- Hydraulic power unit or pressure source required
- More complex piping, leakage control and maintenance

More info: [Rotork hydraulic systems](#)



Selection note: Use hydraulic actuation when required force or torque exceeds practical pneumatic or electric solutions.

Control Valve Accessories

Position Transmitter

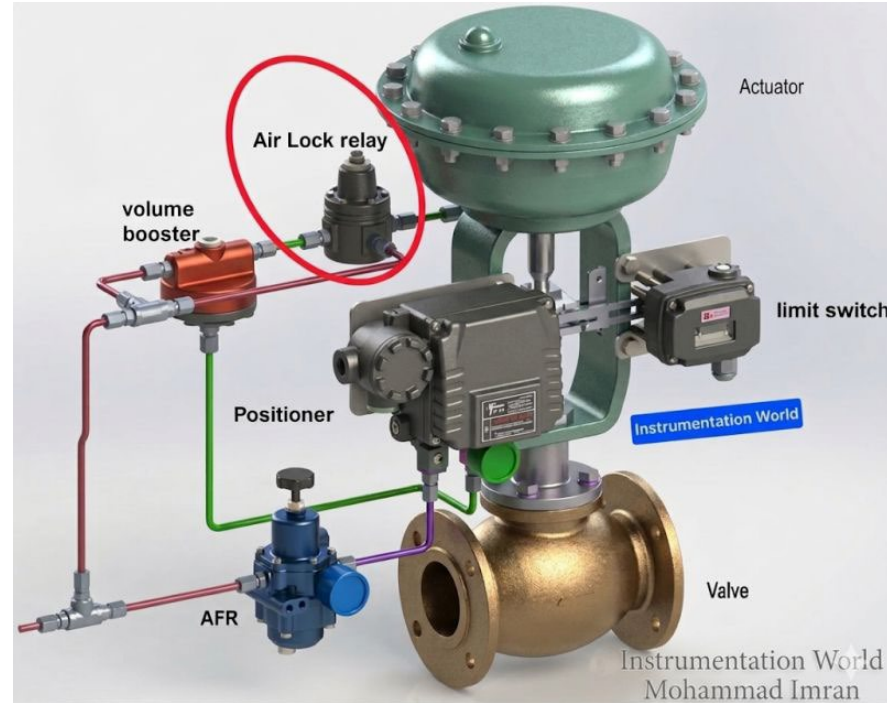
Returns actual valve position for monitoring, feedback and diagnostics.

Limit Switches

Confirm fully open/closed states for interlocks, alarms and sequence control.

Air Filter Regulator (AFR)

Filters compressed air and sets the supply pressure.



Solenoid Valve

Switches actuator air; common in on/off and emergency-shutdown duty.

Manual Override

Enables local operation during maintenance or loss of power/control signal.

Volume Booster (optional)

Raises airflow to large pneumatic actuators for faster response.

SIGNAL FLOW Automation system → Positioner → Actuator | Position transmitter / limit switches → Automation system

AIR SUPPLY Plant air → AFR → Solenoid → Volume booster* → Actuator

Key Point: Performance depends not only on the valve and actuator, but also on accessories used for control, monitoring, safety and diagnostics.

Actuator selection summary

Match valve motion and load first—then utilities, control duty and failure response

Application need	Suitable actuator type	Reason
Common process control valve	Pneumatic diaphragm or cylinder	Fast response and easy positioner integration
Large quarter-turn valve	Scotch yoke or high-output cylinder	High torque, especially near end positions
Location without compressed air	Electric actuator	Only electrical supply is required
Very high force or torque	Hydraulic actuator	High power density for demanding loads
Fail-safe closing required	Spring-return pneumatic or hydraulic	Stored spring energy drives the defined safe state
Precise modulating control	Diaphragm/cylinder with positioner; modulating electric	Controlled positioning and feedback

Learning question

Why does actuator type matter in control-valve selection?

Positioners

Positioner functions

- Receives valve position request
- Measures actual valve position
- Controls actuator pressure
- Minimises position error

Positioner models

- [Neles NDX™ – valve controller](#)
- [Neles™ ND9000™ intelligent valve controller](#)
- [Series 6A - Closer Look | Bray](#)
- [Siemens SIPART PS2 Valve Positioner](#)

Communication

- 4–20 mA + HART
- Profibus PA
- Foundation Fieldbus

Diagnostic perspective

Modern positioners can provide pressure, position and error information to condition monitoring.



Chapter summary and self-assessment

Key points

- A control valve assembly combines the valve, actuator, positioner and accessories
- Valve type, capacity, sizing and gain determine control-loop performance
- Quarter-turn, linear-motion and special-purpose valves suit different duties
- Pneumatic, electric and hydraulic actuators differ in motion, output and utilities
- Acting mode, torque and fail-safe behavior are essential selection criteria
- Positioners and accessories support accuracy, monitoring, safety and diagnostics

Self-assessment questions

- How does a shut-off valve differ from a control valve?
- How do valve capacity and gain affect control performance?
- Which valve type would you choose for accurate throttling—and why?
- How do motion, torque, utilities and safe state guide actuator selection?
- How do single-acting and double-acting actuators differ?
- What do the positioner, position transmitter and limit switches provide?